TRACE: Integration & SLA Feasibility + Model Legal Note

Part 1: Integration & SLA Feasibility

Deployment Options

TRACE uses a hybrid model: **(1) On-device** — Ollama LLM runs on the technician's laptop/tablet; all AI inference is local with no internet required. **(2) Cloud sync** — Supabase syncs offline-queued decisions when reconnected; the back-office approval dashboard reads from Supabase. **(3) Backend API** — FastAPI middleware between frontend and LangGraph agents; deployable on the dealer's local server or as a cloud service.

Estimated OEM Pilot Effort

Integration Point	Effort	Details
OBD-II Scanner API	2 weeks	Connect to Cummins INLINE adapter or ELM327 via serial/Bluetooth.
Dealer Management System	3 weeks	Import repair history and tech rosters from CDK Global / Reynolds & Reynolds.
Warranty System	2 weeks	Push approved repair records to OEM warranty claims system.
SSO / Identity Provider	1 week	Azure AD or Okta integration for authentication.

Total estimated pilot effort: 8 weeks with 2 developers.

SLA Targets

Metric	Target	How Achieved
System Uptime	99.5%	Managed cloud infra; on-device LLM has no external dependency.
Triage Response	<2s (cached) / <30s (cold)	Local LLM inference; caching for repeated fault code patterns.
Full Diagnosis	<60 seconds	Triage + evidence processing + escalation check.
Sync Latency	<5 minutes	Offline queue syncs on reconnection via Supabase real-time.
Data Retention	7 years	Append-only logs per OEM warranty audit requirements.

Sample SLA Contract Excerpt

> *4.1 — **Uptime Guarantee.** Provider guarantees 99.5% monthly uptime for cloud-hosted components (sync service, approval dashboard, API gateway). Scheduled maintenance windows (announced 48h in advance) are excluded.*
>
> *4.2 — **Service Credits.** If Provider fails to meet the 99.5% target, Customer receives a credit of 10% of that month's fees per 0.5% shortfall, up to 30% of the monthly fee. Credits apply to the next billing cycle.*
>
> *4.3 — **On-Device Availability.** The on-device LLM operates independently of cloud services. Core diagnostic functions remain available offline.*

Part 2: Model License & Legal Note

Open-Source AI Model Licenses

	Llama 3.1 8B (Meta)	Mistral 7B (Mistral AI)
License	Meta Llama 3 Community License	Apache License 2.0
Commercial Use	Allowed (<700M MAU)	Unrestricted
Key Restrictions	No training competing models; no "Llama" branding	Standard Apache 2.0 (include license notice)
TRACE Compliance	Fully compliant — weights downloaded via Ollama, not redistributed	Fully compliant

Data Governance & Privacy

1. **No real PII.** All datasets are 100% synthetic. Technician IDs and engineer names are fictional.
2. **GDPR readiness.** Data stays in dealer infrastructure or GDPR-compliant Supabase. On-device LLM means data never leaves the device unless explicitly synced.
3. **Zero external data transmission.** No data sent to Meta, Mistral AI, OpenAI, or any third party.
4. **Audit log integrity.** Append-only logs (no DELETE). Supports ISO 9001 and OEM warranty audits.

Security Controls

1. **Encryption at rest:** SQLite with OS-level AES-256 (BitLocker/FileVault); Supabase AES-256 server-side.

2. **Encryption in transit:** TLS 1.3 for all device-to-cloud communication; HTTPS-only API.
3. **Access controls:** RBAC at the API layer — techs view own tickets, engineers approve escalations, admins manage users. SSO via Azure AD/Okta.
4. **Vulnerability management:** Pinned dependencies monitored via Dependabot; container images scanned with Trivy; LLM model files checksum-verified on download.

Third-Party Libraries

All permissive licenses: LangGraph (MIT), LangChain (MIT), FastAPI (MIT), Streamlit (Apache 2.0), Pydantic (MIT), Ollama (MIT), SQLite (Public Domain). No copyleft (GPL) dependencies.